

IN THE UNITED STATES DISTRICT COURT
FOR THE DISTRICT OF DELAWARE

QORVO, INC.,	)	
	)	
Plaintiff,	)	
	)	
v.	)	C.A. No. 21-1417 (JPM)
	)	
AKOUSTIS TECHNOLOGIES, INC. and	)	DEMAND FOR JURY TRIAL
AKOUSTIS, INC.,	)	
	)	
Defendants.	)	

**PLAINTIFF'S REPLY BRIEF IN SUPPORT OF ITS MOTION FOR SUMMARY
JUDGMENT THAT U.S. PATENT NOS. 7,522,018 AND 9,735,755 ARE NOT INVALID**

OF COUNSEL:

Robert M. Masters
Jonathan R. DeFosse
Timothy P. Cremen
Roy D. Jung
SHEPPARD, MULLIN, RICHTER
& HAMPTON LLP
2099 Pennsylvania Avenue, NW, Suite 100
Washington, DC 20006-6801
(202) 747-1900

Eric K. Gill
Zachary Alper
Theodore Mayer
SHEPPARD, MULLIN, RICHTER
& HAMPTON LLP
12275 El Camino Real, Suite 100
San Diego, CA 92130-4092
(858) 720-8900

James C. Wald
Kazim A. Naqvi
SHEPPARD, MULLIN, RICHTER
& HAMPTON LLP
1901 Avenue of the Stars, 16th Floor
Los Angeles, CA 90067
(310) 228-3700

MORRIS, NICHOLS, ARSHT & TUNNELL LLP
Jack B. Blumenfeld (#1014)
Jeremy A. Tigan (#5239)
Anthony D. Raucci (#5948)
1201 North Market Street
P.O. Box 1347
Wilmington, DE 19899-1347
(302) 658-9200
jblumenfeld@morrisnichols.com
jtigan@morrisnichols.com
araucci@morrisnichols.com

Jennifer Klein Ayers
SHEPPARD, MULLIN, RICHTER
& HAMPTON LLP
2200 Ross Avenue, 20th Floor
Dallas, TX 75201
(469) 391-7400

Attorneys for Plaintiff Qorvo, Inc.

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I. INTRODUCTION

Qorvo moved for summary judgment that Akoustis has not and cannot prove claims 1 and 12 of the '018 Patent and claims 9 and 10 of the '755 Patent (the “Asserted Claims”) are invalid. D.I. 444. More specifically, Qorvo argued that Akoustis cannot meet its burden of showing invalidity by lack of enablement, inadequate written description, indefiniteness, anticipation, or obviousness.

In opposition, Akoustis does not address Qorvo’s arguments on enablement, written description, indefiniteness, or anticipation. As such, Qorvo’s motion for summary judgment is unopposed on those issues and should be granted.

Akoustis argues only that Qorvo’s request for summary judgment of non-obviousness of the Asserted Claims should be denied. But neither Akoustis nor its expert, Dr. Nguyen, has provided the basic factual support required for establishing obviousness and overcoming the Asserted Claims’ presumption of validity. Akoustis fails to utilize the *Graham* factors to support its obviousness assertion and does not provide **any** explanation as to how or why the identified prior art would have been combined by a person of ordinary skill to arrive at the claimed invention. The Federal Circuit has repeatedly stated that this evidence, known as the “motivation to combine,” is a required part of any obviousness analysis. Its omission is fatal to Akoustis’ obviousness position. Summary judgment should be granted in Qorvo’s favor on this issue as well.

II. SUMMARY JUDGMENT SHOULD BE GRANTED ON THE BASES AKOUSTIS HAS NOT OPPOSED

Qorvo requested summary judgment with respect to five distinct invalidity assertions that Akoustis has raised in this case. Specifically, Qorvo argued that Akoustis cannot show the Asserted Claims are invalid under:

- (i) 35 U.S.C. §112 for lack of enablement because Akoustis has not provided evidence of “undue experimentation” (see D.I. 445, §V);

- (ii) 35 U.S.C. §112 for indefiniteness because Akoustis has not provided evidence that any claim element would not be understood with “reasonable certainty” (see D.I. 445, §VI);
- (iii) 35 U.S.C. §112 for lack of written description because Akoustis has not provided evidence that the inventors lacked “possession” of the claimed inventions (see D.I. 445, §VII);
- (iv) 35 U.S.C. §102 as anticipated because Akoustis has not provided evidence that any cited reference discloses element [1c] of claim 1 of the ’018 Patent or elements [9h]/[10h] of the ’755 Patent (see D.I. 445, §VIII); and
- (v) 35 U.S.C. §103 as obvious because Akoustis has not applied the *Graham* factors or provided any evidence of a motivation to combine the prior art (see D.I. 445, §IX).

Akoustis’ Response (D.I. 463) does not dispute that Qorvo is entitled to summary judgment as to requests (i)-(iv), and the Court should grant summary judgment in favor of Qorvo as to these requests. Regarding request (v), for at least the following reasons, Akoustis’ Response does not come close to showing that summary judgment should not be granted to Qorvo.

III. WHAT AKOUSTIS MUST SHOW TO SUPPORT ITS OBVIOUSNESS CONTENTION

Akoustis’ Response does not dispute it has failed to demonstrate that element 1[c] of the ’018 Patent is anticipated by *WO* ’740 (or any other reference) or that elements [9h]/[10h] of the ’755 Patent are anticipated by *DE* ’425 (or any other reference).¹ Instead, Akoustis’ Response falls back to obviousness theories as to those claim elements. Specifically, that: (i) “[*WO* ’740] combined with the background knowledge of a POSITA” (as exemplified by *Zverev* (1967) and *Gerber* (1985)), renders element 1[c] obvious (Resp., 2-6)²; and (ii) *DE* ’425 combined with *Hashimoto* renders elements [9h]/[10h] obvious (Resp., 6-9).³

¹ See D.I. 445 at 8-12 for an explanation of Akoustis’ failures in this regard, to which Akoustis has not responded.

² As discussed below, this basis was not advanced in Akoustis’ *Final Invalidity Contentions*.

³ Akoustis’ Response does not oppose summary judgment of non-obviousness on the remaining obviousness assertions in Dr. Nguyen’s Report (e.g., *US* ’548 combined with *Hashimoto* or *US* ’999 alone). See D.I. 445, p. 13.

To prevail on these obviousness arguments, Akoustis must show that “the claimed invention as a whole would have been obvious to a person of ordinary skill in the art at the time the invention was made.” *Kahn v. Gen. Motors Corp.*, 135 F.3d 1472, 1479 (Fed. Cir. 1998). Invalidity over prior art based on obviousness is ultimately a question of law, but the inquiry rests on factual determinations. *Richardson-Vicks Inc. v. Upjohn Co.*, 122 F.3d 1476, 1479 (Fed. Cir. 1997). These factual determinations, often called the “*Graham* factors, include: (1) the scope and content of the prior art; (2) the differences between the claimed invention and the prior art; (3) the level of ordinary skill in the art; and (4) certain secondary considerations.” *Id.*

Importantly, a determination of obviousness requires not only that each element of the claimed invention can be found in the prior art, but also that (i) an ordinary artisan would have been “motivated to combine the prior art to achieve the claimed invention”; and (ii) “there would have been a reasonable expectation of success in doing so.” *In re Warsaw Orthopedic, Inc.*, 832 F.3d 1327, 1333 (Fed. Cir. 2016); *see also KSR Int’l v. Teleflex Inc.*, 550 U.S. 398, 418 (2007); *Pers. Web Techs., LLC v. Apple, Inc.*, 848 F.3d 987, 991 (Fed. Cir. 2017). This analysis is generally referred to as a showing of a “motivation to combine.”

The motivation to combine analysis must be **specific**. The Federal Circuit has held that a jury cannot reasonably find the motivation to support obviousness based solely on testimony that “is generic and bears no relation to any specific combination of prior art elements,” and that “fails to explain why a person of ordinary skill in the art would have combined elements from specific references in the way the claimed invention does.” *ActiveVideo Networks, Inc. v. Verizon Commc’ns, Inc.*, 694 F.3d 1312, 1327-1328 (Fed. Cir. 2012) (affirming pre-verdict judgment as a matter of law). *See also Securus Techs., Inc. v. Glob. Tel*Link Corp.*, 701 F. App’x 971, 977 (Fed. Cir. 2017) (“Securus failed to explain how or why the skilled artisan would combine the teachings”

and broad characterizations of the prior art “is not enough for Securus to meet its burden of presenting a sufficient rationale to support an obviousness conclusion”); *InTouch Techs., Inc. v. VGO Commc’ns, Inc.*, 751 F.3d 1327, 1353-54 (Fed. Cir. 2014) (the party asserting obviousness must provide a “meaningful explanation for why one of ordinary skill in the art would be motivated to combine [the particular] references at the time of this invention”); and *Innogenetics, N.V. v. Abbott Labs.*, 512 F.3d 1363, 1373 (Fed. Cir. 2008) (“There must be some articulated reasoning with some rational underpinning to support the legal conclusion of obviousness”).

The presence or absence of a motivation to combine references in an obviousness determination is a pure question of fact. *Novartis Pharms. Corp. v. W.-Ward Pharms. Int’l Ltd.*, 923 F.3d 1051, 1059 (Fed. Cir. 2019) (citations and quotation marks omitted). Therefore, summary judgment of non-obviousness is appropriate where the party challenging patent validity fails to show the existence of clear and convincing evidence supporting the factual bases of the legal conclusion of obviousness. *See, e.g., Eisai Co. v. Dr. Reddy’s Labs., Ltd.*, 533 F.3d 1353, 1356, 1358-59 (Fed. Cir. 2008) (affirming district court’s grant of summary judgment of non-obviousness for failure to provide any reason why a skilled artisan would have modified a known compound to arrive at the claimed invention); *see also Mytee Prods., Inc. v. Harris Rsch., Inc.*, 439 F. App’x. 882, 886 (Fed. Cir. 2011) (affirming district court’s grant of summary judgment of non-obviousness for failing to “provide any reason why a person of ordinary skill would have been motivated to combine the references”).

In view of the above, to survive summary judgment on the two bases identified in its Response, Akoustis must come forward with specific evidence that one of ordinary skill in the art would have been motivated to combine: (i) *WO ’740* with the background knowledge of a POSITA (as exemplified by *Zverev* (1967) and *Gerber* (1985)) to arrive at element 1[c] of the ’018 Patent;

and (ii) *DE '425* with *Hashimoto* to arrive at elements [9h]/[10h] of the '755 Patent. As discussed below, Akoustis fails to identify any such evidence. Summary judgement of non-obviousness should therefore be granted in Qorvo's favor.

IV. AKOUSTIS CANNOT SHOW CLAIMS 1 AND 12 OF THE '018 PATENT ARE OBVIOUS

Regarding the '018 Patent, Akoustis cannot dispute that its *Final Invalidity Contentions* fail to advance specific obviousness assertions against claims 1 or 12. As an initial matter, Akoustis only asserted that claims 1 and 12 are "anticipated" (not obvious):

In view of the foregoing, and the indications of anticipation in the chart, above, Akoustis contends that with respect to the '018 Patent, claims **1**, **2**, **12**, and 13 are **anticipated** by at least references A, B, C, and D. Claims 3 and 6 are anticipated by at least reference A. Claims 3, 4, and 5 are obvious in view of at least any of A-D in combination with E, F, and/or G. Claims 10 and 14 are anticipated by at least A and B, and/or are obvious in view of C and/or D in view of A and/or B.⁴

Nor do the charts attached as Exhibit A to Akoustis' *Final Invalidity Contentions* provide any obviousness analysis for claims 1 or 12. *See* D.I. 446, Ex. A at 22-38 (pages A-1 to A-17). Further, no other portion of the *Final Invalidity Contentions* provides any obviousness analysis based on references A-D, such as any analysis of what is missing from these references and some articulated reasoning as to **why** a person of skill would have looked to a second reference to fill in those gaps and **how** that person would have utilized the second reference's teachings in combination with the first's to arrive at the claimed invention.

Akoustis nevertheless argues that it did preserve obviousness contentions against claims 1 and 12 because it included boilerplate language in the *Final Invalidity Contentions*. Resp., 2-4 (citing language in D.I. 446, Ex. A, 10). But such generic boilerplate is deficient as a matter of law because it does not provide the **specific reasoning** required to support the affirmative defense of

⁴ Reference "A" is *WO '740*.

obviousness. *ActiveVideo Networks, Inc. v. Verizon Commc 'ns, Inc.*, 694 F.3d 1312, 1327-1328 (Fed. Cir. 2012) (holding that specific reasons pertaining to specific combinations is required).

Akoustis' Response also argues that the language in the charts attached as Exhibit A to its *Final Invalidity Contentions* referring to what a "skilled artisan would appreciate" from *WO '740* somehow provides an obviousness contention. Resp., 2, 4. Not so. The cited language is simply an allegation of how a person of skill would have understood the disclosures of *WO '740* in connection with Akoustis' assertion of anticipation. Referring to what an ordinary artisan would "appreciate" does not provide any analysis of what is missing from the prior art, nor any articulated reasoning as to **why** a person of skill would have looked to a second reference to fill in that gap, nor **how** the teachings of the second reference would have been combined with the first reference to arrive at the claimed invention. It therefore cannot support an obviousness contention.

As for Dr. Nguyen's *Expert Report*, even if it could properly expand on the bases advanced in the *Final Invalidity Contentions* (it cannot), Akoustis cannot dispute that it fails to advance obviousness assertions against claims 1 and 12 that can survive summary judgment. First, its specific assertions of invalidity do not include **any** obviousness assertions against claim 1 at all. D.I. 446, Ex. B, ¶¶26 (asserting that *WO '740* and *US '171* anticipate claim 1), 60 (same). Nor do the charts attached as Exhibit C to Dr. Nguyen's *Expert Report* provide any obviousness analysis for claim 1. As for claim 12, while Dr. Nguyen's *Expert Report* does nakedly assert that claim 12 is obvious in view of *WO '740* (D.I. 446, Ex. B, ¶¶26, 60), the charts attached as Exhibit C do not provide any obviousness analysis for claim 12.

Akoustis attempts to forestall summary judgment by arguing that "Dr. Nguyen's Invalidity Report shows that claims 1 and 12 of the '018 patent are obvious based upon *WO '740* combined with the background knowledge of a POSITA." Resp., 4. Specifically, Akoustis argues that *WO*

'740 combined with the background art as discussed in *Zverev* (1967) and *Gerber* (1985) render claims 1 and 12 obvious. Resp., 4-5. But these arguments cannot save Akoustis' obviousness assertion. First, neither *Zverev* (1967) nor *Gerber* (1985) is identified in Akoustis' *Final Invalidity Contentions* and cannot now be relied on. Second, even if *Zverev* (1967) and *Gerber* (1985) had been properly identified and could be relied on, Dr. Nguyen's analysis in ¶¶55-59 of his *Expert Report* is simply a serial identification of purported teachings of the various references. There is no explanation therein of what is missing from *WO '740* and some articulated reasoning as to **why** a person of skill would have looked to *Zverev* (1967) and/or *Gerber* (1985) to fill in those gaps and **how** that person would have utilized *Zverev* (1967) and/or *Gerber* (1985)'s teachings in combination with *WO '740*'s to arrive at the claimed invention. Dr. Nguyen's analysis in ¶¶55-59 of his *Expert Report* cannot, therefore, support an obviousness contention.

Finally, Akoustis argues that summary judgment cannot be granted for claims 1 and 12 of the '018 patent because a factual dispute allegedly exists with regard to whether *WO '740* teaches element [1c]. Resp., 3. Not so. Even if there was a factual dispute as to *WO '740*'s teachings (there does not appear to be because Akoustis has not disputed Qorvo's showing that *WO '740* fails to, by itself, anticipate element [1c] at D.I. 445 at 9), this alleged dispute is independent from Akoustis' failure to provide a proper analysis as to why a person of ordinary skill would have been motivated to combine *WO '740* with the background art as discussed in *Zverev* (1967) and *Gerber* (1985). Qorvo should be granted summary judgment due to this failure, regardless if there is some other dispute regarding the teachings of *WO '740* between the parties.

There is no proper obviousness analysis of claims 1 and 12 in either Akoustis' *Final Invalidity Contentions* or Dr. Nguyen's *Expert Report*. For at least this reason, Qorvo is entitled to

summary judgment that Akoustis cannot show these claims are invalid as obvious under 35 U.S.C. §103.

V. AKOUSTIS CANNOT SHOW CLAIMS 9 AND 10 OF THE '755 PATENT ARE OBVIOUS

Regarding the '755 Patent, Dr. Nguyen asserts that: (i) *DE '425* in view of *Hashimoto*; (ii) *US '548* in view of *Hashimoto*; and (iii) *US '999* alone renders obvious claims 9 and 10. D.I. 446, Ex. B, ¶¶26, 90 and Exhibit D to D.I. 446, Ex. B, at 5-9, 17-21, and 31-34.

Qorvo's Opening Brief demonstrates that Dr. Nguyen has failed to undertake a proper obvious analysis for any of these bases because, while he identifies teachings of the various references, he provides no explanation of what is missing from any of *DE '425*, *US '548*, or *US '999* as to any claim element (e.g., elements [9h]/[10h]) and no articulated reasoning as to **why** a person of skill would have looked to a secondary reference (e.g., *Hashimoto*) to fill in those gaps and **how** that person would have utilized the secondary reference's teachings in combination with any of *DE '425*, *US '548*, or *US '999*'s teachings to arrive at the claimed invention.

Akoustis' Response addresses only the purported combination of *DE '425* and *Hashimoto*.⁵ Specifically, Akoustis argues that:

The motivation to combine may be provided by the implicit teachings of *DE '425* and *Hashimoto* and/or the general knowledge of a POSITA. Dr. Nguyen's Invalidity Report explains that *DE '425* and *Hashimoto* both teach the well-known technique of adjusting stack layer thicknesses to modify the real and imaginary parts of a region's propagation constant. (Ex. B ¶ 65.) A genuine dispute of material fact thus exists as to the combination of *DE '425* and *Hashimoto*, precluding summary judgment in this instance.

Resp., 7.

⁵ Akoustis does not assert that the two remaining bases of obviousness in Dr. Nguyen's *Expert Report*: (i) *US '548* in view of *Hashimoto* or (ii) *US '999* alone, should survive summary judgment.

There is no factual dispute that would impact summary judgment. Whether or not a motivation to combine could be provided by *DE '425*, *Hashimoto*, or general POSITA knowledge is not the issue. The issue is whether Dr. Nguyen has provided any analysis as to how *DE '425* is deficient as to, for example, elements [9h]/[10h]), or any articulated reasoning as to **why** a person of skill would have looked to *Hashimoto* to fill in those gaps and **how** that person would have utilized *Hashimoto's* teachings in combination with *DE '425's* teachings to arrive at the claimed invention. No such analysis exists in either the main body of his Expert Report (see ¶¶86-90 that are silent on these issues) or his appended charts (see D.I. 446, Exhibit D to the Expert Report, pp. 5-9 identifying teachings of *DE '425* and *Hashimoto* but providing no analysis for how or why they would be combined).

In view of the above, Akoustis has not shown that Dr. Nguyen has provided the requisite explanation of motivation to support an obviousness contention against claims 9 and 10 of the '755 Patent. Nor can such support be found in Akoustis' *Final Invalidity Contentions*, which do not even specifically identify the combination of *DE '425* and *Hashimoto* or advance anything other than boilerplate motivation language.

VI. CONCLUSION

Based on the foregoing, Qorvo is entitled to summary judgment that the '018 and '755 Patents are not invalid.

MORRIS, NICHOLS, ARSHT & TUNNELL LLP

/s/ Jeremy A. Tigan

OF COUNSEL:

Robert M. Masters
Jonathan R. DeFosse
Timothy P. Cremen
Roy D. Jung
SHEPPARD, MULLIN, RICHTER
& HAMPTON LLP
2099 Pennsylvania Avenue, NW, Suite 100
Washington, DC 20006-6801
(202) 747-1900

Eric K. Gill
Zachary Alper
Theodore Mayer
SHEPPARD, MULLIN, RICHTER
& HAMPTON LLP
12275 El Camino Real, Suite 100
San Diego, CA 92130-4092
(858) 720-8900

James C. Wald
Kazim A. Naqvi
SHEPPARD, MULLIN, RICHTER
& HAMPTON LLP
1901 Avenue of the Stars, 16th Floor
Los Angeles, CA 90067
(310) 228-3700

Jennifer Klein Ayers
SHEPPARD, MULLIN, RICHTER
& HAMPTON LLP
2200 Ross Avenue, 20th Floor
Dallas, TX 75201
(469) 391-7400

February 27, 2024

Jack B. Blumenfeld (#1014)
Jeremy A. Tigan (#5239)
Anthony D. Raucci (#5948)
1201 North Market Street
P.O. Box 1347
Wilmington, DE 19899-1347
(302) 658-9200
jblumenfeld@morrisnichols.com
jtigan@morrisnichols.com
araucci@morrisnichols.com

Attorneys for Plaintiff Qorvo, Inc.

CERTIFICATE OF SERVICE

I hereby certify that on February 27, 2024, I caused the foregoing to be electronically filed with the Clerk of the Court using CM/ECF, which will send notification of such filing to all registered participants.

I further certify that I caused copies of the foregoing document to be served on February 27, 2024, upon the following in the manner indicated:

Stephen B. Brauerman, Esquire
Ronald P. Golden III, Esquire
BAYARD, P.A.
600 North King Street, Suite 400
Wilmington, DE 19801
*Attorneys for Defendants Akoustis
Technologies, Inc. and Akoustis, Inc.*

VIA ELECTRONIC MAIL

Ronald S. Lemieux, Esquire
David S. Elkins, Esquire
Victoria Q. Smith, Esquire
SQUIRE PATTON BOGGS (US) LLP
1841 Page Mill Road, Suite 150
Palo Alto, CA 94304-1216
*Attorneys for Defendants Akoustis
Technologies, Inc. and Akoustis, Inc.*

VIA ELECTRONIC MAIL

/s/ Jeremy A. Tigan

Jeremy A. Tigan (#5239)